

Crab LogPar parameter uncertainties

Controlled comparison: same 44 selected cells; 1D collapses E_{pred} within each N_{hit} bin

Parameter uncertainty	1D N_{hit}	2D $N_{\text{hit}} \times E_{\text{pred}}$	Reduction
Normalization σ_{ϕ_0}	4.318×10^{-14}	3.461×10^{-14}	19.8%
Spectral index σ_{α}	0.02317	0.02048	11.6%
Curvature σ_{β}	0.01693	0.01358	19.8%

Formal HESSE 1σ uncertainties at pivot energy $E_0 = 3 \text{ TeV}$. ϕ_0 uncertainty unit: $\text{TeV}^{-1} \text{ cm}^{-2} \text{ s}^{-1}$.

Lower uncertainty is better; this table quantifies formal statistical precision, not bias or systematic uncertainty.